Math 110 — Week 5

Solutions

HW5 P1.

Idea. The null space is given by two zeros, so factor. Surjectivity: find polynomials sent to e_1 and e_2. Those same polynomials give the basis of the quotient.

1. (a) $p \in \operatorname{null} T$ means $p(0) = p(1) = 0$, i.e. $x(x-1)$ divides p. With $\deg p \le 3$: $p = x(x-1)(ax+b)$. So $\operatorname{null} T = \operatorname{span}\big(x(x-1),\ x^2(x-1)\big)$, of dimension 2.
2. $T(1-x) = (1,0)$ and $T(x) = (0,1)$. These span $\mathbb{R}^2$, so T is surjective. (Check: $4 = 2+2$.)
3. (b) $\widetilde{T}\colon \mathcal{P}_3(\mathbb{R})/\operatorname{null} T \to \mathbb{R}^2$, $\widetilde{T}(p + \operatorname{null} T) = (p(0), p(1))$. It is well-defined because two names of the same set differ by a polynomial vanishing at 0 and 1.
4. (c) $\widetilde{T}$ is an isomorphism, and $\widetilde{T}\big((1-x) + \operatorname{null} T\big) = e_1$, $\widetilde{T}\big(x + \operatorname{null} T\big) = e_2$. The inverse of an isomorphism sends a basis to a basis, so $(1-x) + \operatorname{null} T,\ x + \operatorname{null} T$ is a basis of the quotient.

HW5 P2.

Idea. The formula is forced, so uniqueness is immediate. Existence is the well-definedness check, and it uses exactly $U \subseteq \operatorname{null} T$.

1. *Uniqueness.* Every element of V/U is $v + U$ for some v, and the condition prescribes the value Tv there. So at most one such map exists.
2. *Well-defined.* If $v + U = w + U$, then $v - w \in U \subseteq \operatorname{null} T$, so $Tv - Tw = T(v-w) = 0$.
3. *Linear.* $\widetilde{T}(\lambda(v+U) + (w+U)) = \widetilde{T}((\lambda v + w) + U) = T(\lambda v + w) = \lambda\widetilde{T}(v+U) + \widetilde{T}(w+U)$.
4. *Null space.* $\widetilde{T}(v+U) = 0 \iff Tv = 0 \iff v \in \operatorname{null} T$. So $\operatorname{null}\widetilde{T} = \{v + U : v \in \operatorname{null} T\} = (\operatorname{null} T)/U$.

HW5 P3.

Idea. If φ takes one nonzero value, scaling the input produces every value.

1. Suppose $\varphi \neq 0$. Choose v with $\varphi(v) = c \neq 0$.
2. For any $a \in \mathbb{F}$: $\varphi\big(\frac{a}{c}v\big) = \frac{a}{c}\varphi(v) = a$. So φ is surjective.

HW5 P4.

Idea. φ_j reads off the j-th coordinate. Functionals are compared by evaluating on the basis.

1. (a) Write $v = \sum_k a_k v_k$. Then $\varphi_j(v) = \sum_k a_k \varphi_j(v_k) = a_j$. Substituting $a_j = \varphi_j(v)$ gives the formula.
2. (b) Evaluate the right side at v_k: $\sum_j \psi(v_j)\varphi_j(v_k) = \psi(v_k)$. So the two functionals ψ and $\sum_j \psi(v_j)\varphi_j$ agree on a basis of V, hence are equal.

HW5 P5.

Idea. Column j of $\mathcal{M}(T')$ is the coordinate vector of $T'(w_j')$, and coordinates of a functional are its values on $v_1, \dots, v_n$.

1. Let $B = (v_1, \dots, v_n)$, $C = (w_1, \dots, w_m)$, $A = (a_{lk})$, so $Tv_k = \sum_l a_{lk} w_l$.
2. $T'(w_j')(v_k) = w_j'(Tv_k) = w_j'\big(\sum_l a_{lk} w_l\big) = a_{jk}$.
3. By P4(b), $T'(w_j') = \sum_k T'(w_j')(v_k)\, v_k' = \sum_k a_{jk} v_k'$.
4. So the (k,j) entry of $\mathcal{M}(T')$ is a_{jk}: $\mathcal{M}(T') = A^{\mathsf{T}}$.

HW5 P6.

Idea. A row is in U^0 when it gives 0 on each of the two spanning vectors.

1. $(a\ b\ c)$ applied to $(1,1,0)$ is $a+b$; applied to $(0,1,1)$ is $b+c$. Both zero: $a=-b$, $c=-b$.
2. $U^0 = \operatorname{span}\big((1\ \ -1\ \ 1)\big)$, the functional $(x,y,z)\mapsto x-y+z$.
3. $\dim U = 2$ (the two vectors are not multiples of each other), $\dim U^0 = 1$, and $2+1=3$.

HW5 P7.

Idea. Injective: for $v\neq 0$ build a functional with $\varphi(v)=1$ by making v a basis vector.

1. j_V is linear: $j_V(\lambda v+w)(\varphi) = \varphi(\lambda v+w) = \lambda\varphi(v)+\varphi(w)$.
2. Let $v\neq 0$. Extend v to a basis of V and let φ be 1 on v and 0 on the other basis vectors. Then $j_V(v)(\varphi)=1$, so $j_V(v)\neq 0$. Hence $\operatorname{null} j_V = \{0\}$.
3. If $\dim V = n < \infty$: $\dim V'' = \dim V' = n$. An injective map between spaces of the same finite dimension is an isomorphism.

R1.

Idea. Same map as HW5 P1.

1. (a) $E(\lambda p+q) = (\lambda p(0)+q(0), \lambda p(1)+q(1)) = \lambda E(p)+E(q)$. $E(1-x)=(1,0)$, $E(x)=(0,1)$: surjective.
2. (b) $U=\{x(x-1)(ax+b)\}$, basis $x(x-1)$, $x^2(x-1)$ (different degrees, and $\dim U = 4-2=2$).
3. (c) Well-defined: if $p+U=q+U$ then $p-q\in U=\operatorname{null} E$, so $E(p)=E(q)$. Linear because E is. Injective: $\widetilde{E}(p+U)=0 \Rightarrow p\in U \Rightarrow p+U=0+U$. Surjective because E is.
4. (d) $(1-x)+U\mapsto(1,0)$ and $x+U\mapsto(0,1)$. These two elements form a basis of V/U because their images form a basis and $\widetilde{E}$ is an isomorphism.

R2.

Idea. Everything follows from $(T^*\varphi)(v)=\varphi(Tv)$.

1. (a) $T^*(\varphi)=\varphi\circ T$ for $\varphi\in W^*$. Linear: $T^*(\lambda\varphi+\psi)=(\lambda\varphi+\psi)\circ T=\lambda(\varphi\circ T)+\psi\circ T$.
2. (b) For $\varphi\in Z^*$: $(S\circ T)^*\varphi = \varphi\circ S\circ T = (S^*\varphi)\circ T = T^*(S^*\varphi)$.
3. (c) Suppose $T^*\varphi=0$, i.e. $\varphi(Tv)=0$ for all v. Since T is surjective, every $w\in W$ is some Tv, so $\varphi(w)=0$ for all w. So $\varphi=0$, and T^* is injective.
4. (d) $(T^*\ell)(x,y)=\ell(2x-y)=6x-3y$.

E1.

Idea. Rows of the inverse of the matrix whose columns are the basis.

1. $M=\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$, $\det M=-2$, $M^{-1}=\frac{1}{-2}\left(\begin{smallmatrix}-1 & -1\\ -1 & 1\end{smallmatrix}\right)=\left(\begin{smallmatrix}1/2 & 1/2\\ 1/2 & -1/2\end{smallmatrix}\right)$.
2. $\varphi_1=(\frac{1}{2}\ \ \frac{1}{2})$, $\varphi_2=(\frac{1}{2}\ \ -\frac{1}{2})$. Check: $\varphi_1(v_1)=1$, $\varphi_1(v_2)=0$, $\varphi_2(v_1)=0$, $\varphi_2(v_2)=1$.
3. $\psi=\psi(v_1)\varphi_1+\psi(v_2)\varphi_2 = 4\varphi_1+2\varphi_2$. Check: $4(\frac{1}{2}\ \ \frac{1}{2})+2(\frac{1}{2}\ \ -\frac{1}{2})=(3\ \ 1)$.